Best Deep Learning Models for Inference with Stock Options Market Data

Long Short-Term Memory (LSTM)

LSTMs are effective for capturing long-term dependencies in time series data, such as stock prices. They are designed to avoid the vanishing gradient problem that can occur with traditional RNNs, allowing them to capture long-term dependencies in sequential data. An LSTM could be used for predicting stock prices based on historical data. However, LSTMs require a significant amount of data to train effectively and can be computationally intensive. Overfitting can also be an issue.

Gated Recurrent Unit (GRU)

GRUs are similar to LSTMs but with fewer parameters, making them faster and efficient for certain datasets. A GRU could be used for real-time sentiment analysis of tweets related to a particular stock. However, GRUs may not be as powerful as LSTMs in capturing long-term dependencies and can also be prone to overfitting, especially with limited data.

Convolutional Neural Networks (CNN)

CNNs can efficiently process and learn from spatial data, which can be beneficial for identifying patterns in tick-by-tick stock market data. A CNN could be used to analyze tick-by-tick stock market data. However, CNNs may require a significant amount of data to train effectively and can be computationally expensive. They may not be as effective for purely sequential data without spatial components.

Recurrent Neural Network (RNN)

RNNs are designed to work with sequences of data and are capable of learning temporal dynamics. However, traditional RNNs suffer from the vanishing gradient problem, making them less effective for long sequences. An RNN could be used for predicting the next day's opening price of a stock based on the previous days' closing prices. Traditional RNNs can struggle with learning long-term dependencies due to the vanishing gradient problem. They can also be computationally expensive and difficult to train.

Transformers

Transformers are a type of neural network architecture that uses self-attention mechanisms to learn dependencies between different elements in a sequence. They have been highly successful in various natural language processing tasks. A transformer could be used for high-frequency trading, where it would analyze the relationships between different stocks and predict the best trading opportunities based on historical data. However, transformers can be computationally expensive and require a significant amount of data to train effectively. They may also struggle with tasks that require understanding long-term dependencies in sequential data.

Time used: 131.40 seconds

Token usage: LM Studio | model: openai/phi-4-mini-instruct

Token Usage Details

Per LLM Call

Step	Input Tokens	Output Tokens	Total Tokens
Selector	86	349	435
Explainer	398	717	1115
Formatter	1157	583	1740

Totals

Prompt Tokens	Completion Tokens	Total Tokens
1641	1649	3290

Performance

Elapsed (s)	Completion tok/s	Total tok/s
131.4	12.55	25.04